Dextromethorphan PBPK model: fit PK

1 Methods: Dextromethorphan PBPK model: fit PK

1.1 Parameters and data

13 parameters were optimized: Ka_dis [1/hr] dissolution dextromethorphan (`Ka_dis_dxm`), fraction absorbed dextromethorphan (`GU__F_dxm_abs`), Vmax DXMCYP2D6 (`LI__DXMCYP2D6_Vmax`), Vmax DXMCYPX (`LI__DXMCYPX_Vmax`), Vmax DXMNDM (`LI__DXMNDM_Vmax`), Vmax DXONDM (`LI__DXONDM_Vmax`), Vmax DXOUGT (`LI__DXOUGT_Vmax`), tissue/plasma partition coefficient dxm (`Kp_dxm`), tissue/plasma partition coefficient mm (`Kp_mm`), renal clearance of unbound dextromethorphan relative to the GFR (`KI__DXMEX_f`), renal clearance of unbound dextrorphan relative to the GFR (`KI__DXOEX_f`), renal clearance of unbound dextrorphan glucuronide relative to the GFR (`KI__DXOGLUEX_f`) and renal clearance of unbound N-demethylated morphinans relative to the GFR (`KI__MMEX_f`).

A fit mapping is a curve of a study, the data and the observable of the model it is compared with. The fit has 227 fit mappings of training data with 2165 data points, whose residuals enter the objective; 362 fit mappings of validation data with 2526 data points, which were simulated and evaluated with the fitted parameters but not fitted; and 79 fit mappings of outliers with 408 data points, data the fit dropped as not usable, which were evaluated like the validation data so that the decision to drop them can be checked. The fit was set up and run with sbmlsim [1].

1.2 Objective function

The parameters were estimated by minimizing the objective function

$$F(\vec{p}) = \frac{1}{2} \sum_k \sum_i \rho\left(w_{i,k} r_{i,k}(\vec{p})^2\right) \tag{1}$$

where k runs over the fit mappings of the training data and i over their data points, $r_{i,k}$ is the residual, $w_{i,k} = w_k w^p_{i,k}$ the weight of the data point and ρ the loss function, applied to the squared weighted residual $z = w_{i,k} r^2_{i,k}$. The loss function is linear, $\rho(z) = z$, i.e. a standard least squares problem.

The residuals are normalized by the mean $\overline{y}_k$ of the data of their curve, which makes curves of different quantities and magnitudes comparable:

$$r_{i,k} = \frac{y_{i,k} - f_{k(x_{i,k},\vec{p})}}{\overline{y}_k} \tag{2}$$

The weight of curve k is $w_k = \frac{n_k}{N_k}$, the weight n_k of its fit mapping (the weight its fit mapping collection assigns, 1 by default, or, for a collection which uses the weights of its mappings, the weight of the mapping, by default the number of subjects of its data) divided by the number N_k of its data points.

A data point is weighted with the inverse of its coefficient of variation, $w^p_{i,k} = |y_{i,k} \frac{|}{\sigma_{i,k}}$, with $\sigma_{i,k}$ the standard deviation of the data point, or its standard error for data without a standard deviation. An error of zero is missing, a missing error of a curve with errors is replaced by the largest error of the curve, and the data points of a curve without errors have the weight $w^p_{i,k} = 2$, the weight of a coefficient of variation of 0.5.

1.3 Optimization

The optimizer searched the parameters in log10 space within their bounds; bounds, values and intervals are given in the units of the fit parameters, the units of the table of the parameters.

The objective was minimized in 3 runs with differential evolution [2] of SciPy [3] within the bounds of the parameters, with the strategy `best1bin`, a population of 15 times the number of parameters, at most 1000 generations, the relative tolerance 0.01 of convergence, the recombination constant 0.7 and a mutation dithered between 0.5 and 1. The initial population of a run was drawn by latin hypercube sampling [4], with a seed of every run derived from the seed 1234 of the fit. The best member of the final population was polished with the L-BFGS-B method of SciPy.

The convergence of the runs was assessed from the traces of their cost and the waterfall plot of their final costs, and the parameter set with the lowest cost was used.

The models were simulated with libRoadRunner [5] and its integrator CVODE [6] with the relative tolerance 10^{-6} and the absolute tolerances 10^{-13} of an amount, 10^{-13} of a concentration and 10^{-6} of the other states, which for a species in a concentration is multiplied by the initial volume of its compartment. It ran with the variable step size of

libRoadRunner, i.e. with an output at every step of the integrator. A data point at the time of a change after the start of a simulation, e.g. a dose or the reset of a collection, was compared with the state before the change, except the first data point of a curve with data at later times, which starts a collection and was compared with the state after the change.

1.4 Goodness of fit

The goodness of fit is given per kind of data (training data, validation data and outliers) by metrics over its n data points with the residuals $e_i = y_i - f\left(x_i, \hat{\vec{p}}\right)$ at the fitted parameters $\hat{\vec{p}}$, the mean squared error, its root and the root mean squared error normalized by the mean $\overline{y}_{k(i)}$ of the data of the curve of a point:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n} e_i^2, \quad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}, \quad \mathrm{NRMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_i}{\overline{y}_{k(i)}}\right)^2} \tag{3}$$

The root mean square of the residuals of the objective, i.e. what the fit minimizes, and the coefficient of determination:

$$\mathrm{RMSE}_w = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\rho(w_i r_i^2)}, \quad R^2 = 1 - \frac{\sum_i e_i^2}{\sum_i \left(y_i - \overline{y}\right)^2} \tag{4}$$

The predictions of a non-linear model are no linear regression of the data, so R^2 can be negative: the prediction is worse than the mean of the data. The information criteria of Akaike [7] and Schwarz [8] with the number n_p of fitted parameters are those of a least squares fit with normally distributed residuals, up to an additive constant, so only differences between models fitted to the same data are meaningful:

$$\mathrm{AIC} = n\ln(\mathrm{MSE}) + 2n_p, \quad \mathrm{BIC} = n\ln(\mathrm{MSE}) + n_p\ln(n) \tag{5}$$

The percentage of the data points whose prediction $f_i = f\left(x_i, \hat{\vec{p}}\right)$ lies within twofold of the data is taken over the n_+ data points with positive data $y_i > 0$ and a positive prediction $f_i > 0$; the others have no ratio and are not counted, and a kind of data without such a point has no percentage:

$$P_2 = \frac{100}{n_+}\left|\left\{i : \frac{1}{2} \le \frac{f_i}{y_i} \le 2\right\}\right| \tag{6}$$

The goodness of fit plot shows the predictions against the data on logarithmic axes, with the lines of twofold $f = 2y$ and $f = \frac{y}{2}$. The Bland-Altman plot [9] shows the ratio of prediction and data as $d_i = \log_{10}\left(\frac{f_i}{y_i}\right)$ against the geometric mean $\sqrt{f_i y_i}$ for the data points with positive values, with the bias $\overline{d}$ and the limits of agreement $\overline{d} \pm 1.96 s_d$, with the sample standard deviation s_d of d_i, of the training data; they are read as the fold factors $10^{\overline{d}}$ and $10^{\overline{d} \pm 1.96 s_d}$.

1.5 Identifiability

The identifiability of the parameters was analysed with the profile likelihood [10], [11]. The profile of a parameter is the lowest objective with the parameter fixed:

$$\mathrm{PL}(p_j) = \min_{p_{l \ne j}} F(\vec{p}) \tag{7}$$

At every point of a scan the other parameters were re-optimized from the previous point with the least squares optimizer of SciPy with the relative step 0.05. A scan ran from the optimum in both directions in the space of the optimizer, with steps in decades of the parameter, starting at 0.1. A step which raised the objective by more than 0.2 of the distance between the objective at the optimum and the threshold was reduced by the factor 2 and repeated, down to 0.01, and the step after one which raised it by less than that divided by the factor was enlarged by the factor, up to 1. A scan stopped at the threshold, at the bound of the parameter, at a point which failed or after 50 points.

The confidence intervals are the values at which the profile crosses the threshold of the likelihood ratio test

$$F_{\mathrm{thr}} = F_{\mathrm{min}} + \chi^2_\alpha \frac{\mathrm{df}}{2} \tag{8}$$

with F_{min} the lowest objective of the fit and of all profiles and $\chi^2_\alpha(\mathrm{df})$ the quantile of the χ^2 distribution at the confidence level $\alpha = 0.95$ with df = 1 degrees of freedom, i.e. 1.92 above the lowest objective. The threshold has this confidence level if F is the negative log-likelihood of the data up to a constant, i.e. if the weighted residuals are independent and normally distributed with unit variance; otherwise the level is approximate. A crossing is

interpolated between two points of the profile, on the logarithm of a parameter on a logarithmic scale. A parameter is identifiable if its profile reaches the threshold within the scan on both sides of the optimum, i.e. its confidence interval is finite; practically non-identifiable towards the lower bound if its profile reaches the threshold above the optimum but not below it within the scan, which ends at the lower bound or after 50 points; practically non-identifiable towards the upper bound if its profile reaches the threshold below the optimum but not above it within the scan, which ends at the upper bound or after 50 points; practically non-identifiable if its profile reaches the threshold on neither side within the scans, which end at the bounds or after 50 points, and is not flat; and structurally non-identifiable if its profile is flat on both sides. A side of a profile is flat if it rises by less than 0.05 of the distance between the lowest objective and the threshold, and a profile with a point which failed or did not converge is not classified. The profile likelihood is preferred over the intervals of the Fisher information, its intervals are invariant under a transformation of the parameters and can be asymmetric [12].

The Fisher information is the local alternative to the profiles: it is the Gauss-Newton approximation of the curvature of the objective at the optimum, from the Jacobian J of the weighted residuals of the training data with respect to the parameters $\vec{p}$ in the space of the optimizer, computed by central finite differences with the step $h_j = \varepsilon \max\left(|p_j|, 1\right),\ \varepsilon = 10^{-6}$ of the parameter p_j:

$$I = J^\top J, \quad \mathrm{cov} = \sigma^2 I^{-1}, \quad \sigma^2 = \frac{2F}{n - n_p} \tag{9}$$

with the number n of data points of the training data and the number n_p of parameters. The information has rank 13 of 13, the number of its eigenvalues above 10^{-8} of the largest one, and the condition number 515; the covariance of an information of a lower rank is computed with the pseudo-inverse. The Wald intervals are $\hat{p}_j \pm t_{\frac{1+\alpha}{2}, n-n_p} \mathrm{SE}_j$ with the standard errors $\mathrm{SE}_j = \sqrt{\mathrm{cov}_{jj}}$ at the confidence level $\alpha = 0.95$, transformed back into the units of the fit parameters.

1.6 Provenance

The fit PK_20261020_094137__4239e was run on 2026-10-20 07:41:49 UTC with sbmlsim 0.10.0 [1], libRoadRunner 2.10.0 [5], SciPy 1.18.1 [3], NumPy 2.5.3 [13] and Python 3.14.4 (seed 1234). The models and data were those of commit `aa8deb245665a611f7c5be43a35ced599c893cf3` of `/home/mkoenig/git/pkdb_models-dextromethorphan` their SHA-256 checksums are listed in Table 1.

Table 1: SHA-256 checksums of the models and data of the fit PK_20261020_094137__4239e, their first 12 characters; the full checksums are in the provenance of the result (optimization_result.json).

File	SHA-256
pkdb_models/models/dextromethorphan/data/caffeine/Armani2017/.Armani2017_Fig1.tsv	1afb6f9c2167...
pkdb_models/models/dextromethorphan/data/caffeine/Armani2017/.Armani2017_Fig2.tsv	2eca5f2f6f8c...
pkdb_models/models/dextromethorphan/data/caffeine/Armani2017/.Armani2017_Tab1.tsv	b99e9c447417...
pkdb_models/models/dextromethorphan/data/caffeine/Armani2017/.Armani2017_TabGroups.tsv	e2ae763498cb...
pkdb_models/models/dextromethorphan/data/caffeine/Armani2017/.Armani2017_TabInterventions.tsv	6219415a1ec4...
pkdb_models/models/dextromethorphan/data/caffeine/Doroshyenko2013/.Doroshyenko2013_Fig1.tsv	5cd220216822...
pkdb_models/models/dextromethorphan/data/caffeine/Doroshyenko2013/.Doroshyenko2013_Tab1.tsv	9c9dc559925a...
pkdb_models/models/dextromethorphan/data/caffeine/Doroshyenko2013/.Doroshyenko2013_Tab2.tsv	e3b0c44298fc...
pkdb_models/models/dextromethorphan/data/caffeine/Doroshyenko2013/.Doroshyenko2013_TabData.tsv	4d0dba3d8250...
pkdb_models/models/dextromethorphan/data/caffeine/Oh2012/.Oh2012_Fig3.tsv	e60ef89928e7...
pkdb_models/models/dextromethorphan/data/caffeine/Oh2012/.Oh2012_TabGroups.tsv	021f55009aad...

File	SHA-256
pkdb_models/models/dextromethorphan/data/caffeine/Oh2012/.Oh2012_TabInterventions.tsv	2ccc461e26f0...
pkdb_models/models/dextromethorphan/data/caffeine/Yamazaki2016/.Yamazaki2016_Fig2.tsv	b431b1f2bcf3...
pkdb_models/models/dextromethorphan/data/caffeine/Yamazaki2016/.Yamazaki2016_Fig3.tsv	5e1d7ec4d97e...
pkdb_models/models/dextromethorphan/data/caffeine/Yamazaki2016/.Yamazaki2016_Tab1.tsv	415008aace6c...
pkdb_models/models/dextromethorphan/data/caffeine/Yamazaki2016/.Yamazaki2016_TabGroups.tsv	0df0d66858f8...
pkdb_models/models/dextromethorphan/data/caffeine/Yamazaki2016/.Yamazaki2016_TabInterventions.tsv	c47ac863906b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Abdelrahman1999/.Abdelrahman1999_Tab1.tsv	1a1f7c06cc53...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Abdelrahman1999/.Abdelrahman1999_TabIndividuals.tsv	3b420213adf7...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Abdelrahman1999/.Abdelrahman1999_TabInterventions.tsv	93e303e48413...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Abduljalil2010/.Abduljalil2010_Fig3.tsv	f880fb3f5868...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Abduljalil2010/.Abduljalil2010_TabIndividuals.tsv	698d925fefe1...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Abduljalil2010/.Abduljalil2010_TabInterventions.tsv	8764be831d1c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Abdulmanap1999/.Abdulmanap1999_Tab1.tsv	ac439e10b211...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Abdulmanap1999/.Abdulmanap1999_TabInterventions.tsv	94938bffe2c7...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Barnhart1980/.Barnhart1980_Fig4.tsv	2308fc7cfdd4...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Barnhart1980/.Barnhart1980_Fig5.tsv	b4d22e118968...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Barnhart1980/.Barnhart1980_Fig6.tsv	4c09852debe5...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Barnhart1980/.Barnhart1980_Tab2.tsv	cda69241a22c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Barnhart1980/.Barnhart1980_TabGroups.tsv	25a9234cffb8...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Barnhart1980/.Barnhart1980_TabIndividuals.tsv	2df26dab2c64...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Barnhart1980/.Barnhart1980_TabInterventions.tsv	e4a24bc390d8...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Capon1996/.Capon1996_Fig2.tsv	03c4b97f8aad...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Capon1996/.Capon1996_Fig3.tsv	a8d61eb1ee1b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Capon1996/.Capon1996_Fig4.tsv	d78ed666d91e...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Capon1996/.Capon1996_Fig5.tsv	f965aa773eca...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Capon1996/.Capon1996_Tab1.tsv	4b3241337f76...

File	SHA-256
pkdb_models/models/dextromethorphan/data/dextromethorphan/Capon1996/.Capon1996_Tab2.tsv	005fd131b784...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Capon1996/.Capon1996_TabA.tsv	44dc6f1dc821...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Capon1996/.Capon1996_TabIndividuals.tsv	44dc6f1dc821...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chen2017/.Chen2017_Fig2A.tsv	d93b12c26698...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chen2017/.Chen2017_TabData.tsv	ee83dced4d50...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chen2017/.Chen2017_TabGroups.tsv	cded39c6c261...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chen2017/.Chen2017_TabInterventions.tsv	d2cc50626969...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chladek2000/.Chladek2000_Fig3.tsv	651590b36248...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chladek2000/.Chladek2000_Fig3A.tsv	26007ab609da...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chladek2000/.Chladek2000_Fig3B.tsv	51595d8599fa...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chladek2000/.Chladek2000_Fig4A.tsv	d85949a35530...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chladek2000/.Chladek2000_Fig4B.tsv	78a5ed29d465...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chladek2000/.Chladek2000_Tab1.tsv	c3ce5633c29d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chladek2000/.Chladek2000_Tab2.tsv	97e905e25d52...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chladek2000/.Chladek2000_TabData.tsv	b170eccb0570...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chladek2000/.Chladek2000_TabGroups.tsv	d4b307b4eec3...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chladek2000/.Chladek2000_TabIndividuals.tsv	a5217d63a8d5...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Chladek2000/.Chladek2000_TabInterventions.tsv	eeca45afc4ce...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Demirbas1998/.Demirbas1998_Fig2.tsv	8a669eded783...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Demirbas1998/.Demirbas1998_Tab1.tsv	61481b2d6b05...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Demirbas1998/.Demirbas1998_Tab3.tsv	eb4c30f7489c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Dorado2017/.Dorado2017_Fig1.tsv	f38cca0a92cd...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Dorado2017/.Dorado2017_Fig2.tsv	a4d0a24930e0...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Dorado2017/.Dorado2017_Fig3.tsv	1baffde64ef2...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Dorado2017/.Dorado2017_Tab2Groups.tsv	933c52223b8f...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Dorado2017/.Dorado2017_Tab2Individuals.tsv	249ab4dfdde1...

File	SHA-256
pkdb_models/models/dextromethorphan/data/dextromethorphan/Dorado2017/.Dorado2017_TabGroups.tsv	8ec5b7d7ceda...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Dorado2017/.Dorado2017_TabIndividuals.tsv	15c22b907c2d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Dorado2017/.Dorado2017_TabInterventions.tsv	100e1450071f...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Droll1998/.Droll1998_Fig1.tsv	e985739031df...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Droll1998/.Droll1998_Fig2.tsv	fe0942a2eba7...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Droll1998/.Droll1998_Fig2_3D_todo.tsv	1dccbc129f83...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Droll1998/.Droll1998_Fig2_todo.tsv	1dccbc129f83...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Droll1998/.Droll1998_Tab1.tsv	110029c0f58a...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Droll1998/.Droll1998_TabIndividuals.tsv	a2a8200be5d6...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Duedahl2005/.Duedahl2005_Fig3.tsv	cea74060843d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Duedahl2005/.Duedahl2005_Tab2.tsv	42a99b0a3139...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Duedahl2005/.Duedahl2005_TabGroups.tsv	9d8acefd702a...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Duedahl2005/.Duedahl2005_TabInterventions.tsv	971a202e862e...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Dumond2010/.Dumond2010_Fig1.tsv	307ad0dbf823...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Dumond2010/.Dumond2010_Tab1.tsv	3e2141c4a765...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Edwards2017/.Edwards2017_Fig1.tsv	f4a652963e25...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Edwards2017/.Edwards2017_Fig2.tsv	e1e452ccecda...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Edwards2017/.Edwards2017_Fig3.tsv	93012ed5baa0...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Edwards2017/.Edwards2017_Fig4.tsv	352a8d5bb812...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Edwards2017/.Edwards2017_Fig5.tsv	49e70b690281...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Edwards2017/.Edwards2017_Fig6.tsv	83ec0dfb937c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Edwards2017/.Edwards2017_Fig7.tsv	d04c9970473c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Edwards2017/.Edwards2017_Tab2.tsv	71f7b2a5ab7d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Edwards2017/.Edwards2017_Tab3.tsv	c2a9e7cf50e1...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Edwards2017/.Edwards2017_TabA.tsv	843a06e036b9...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Edwards2017/.Edwards2017_TabGroups.tsv	71f7b2a5ab7d...

File	SHA-256
pkdb_models/models/dextromethorphan/data/dextromethorphan/Eichhold1997/.Eichhold1997_Fig6.tsv	627649758632...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Eichhold2007/.Eichhold2007_Fig4.tsv	b23e2286a3d6...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Eichhold2007/.Eichhold2007_Fig5.tsv	40752f07fe21...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab100.tsv	4e3725dcfb78...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab101.tsv	7331e46540f8...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab102.tsv	831cb9819def...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab103.tsv	12ab15bccacc...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab104.tsv	c51d6da266d2...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab105.tsv	d896d531f6a9...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab106.tsv	183b96a8f134...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab108.tsv	b58608d14f82...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab109.tsv	5858a3370b2c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab110.tsv	0666f1c9dde1...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab111.tsv	b04b900e7e7d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab112.tsv	b52dd5713ed7...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab113.tsv	e0cd4a1b2dfb...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab114.tsv	20a2e3062487...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab116.tsv	ced20a27c8d6...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab117.tsv	56b8e3bd19b4...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab119.tsv	ee7625544307...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab120.tsv	0770c121c38a...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab121.tsv	5c53c6f79484...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab123.tsv	77f14c1e3785...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab126.tsv	d176033c2447...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab127.tsv	1b8e7f48effc...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab129.tsv	4a0c89d0357e...

File	SHA-256
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab130.tsv	a6e14e6ad137...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab131.tsv	6a9834e77e13...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab134.tsv	b6f71626b354...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab135.tsv	7fe78eebf394...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab137.tsv	413231d96ed3...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab139.tsv	06d223a08c4c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab141.tsv	27a81980bb4b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab144.tsv	6e11e033e834...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab145.tsv	86631c1a472b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab147.tsv	6c7631b1f411...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab148.tsv	c1b7d1eb9588...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab149.tsv	fcabe85be14c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab152.tsv	ee00489866a6...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab153.tsv	c7dd1a1c7002...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab155.tsv	1ebb0f4351b4...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab157.tsv	03b1a8d0cbe3...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab159.tsv	fb9a174dcf85...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab161.tsv	42dca0593500...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab162.tsv	be522b059f4c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab163.tsv	a1a8c918e7b2...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab82.tsv	7e53f1ecec3c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab83.tsv	30cfa4396802...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab84.tsv	2df544f827e1...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab85.tsv	12ef1c063da3...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab86.tsv	789d3674f2eb...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab87.tsv	c6be56f7c02e...

File	SHA-256
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab88.tsv	d59bfc91e55d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab90.tsv	6c356f9d29ce...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab91.tsv	75675f359cb4...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab92.tsv	9810d286cbaa...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab93.tsv	4530c64d0d9b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab94.tsv	35161ca10b2b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab95.tsv	4b4659422901...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab96.tsv	846484e4330e...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab98.tsv	17c20cccfa5c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_Tab99.tsv	5c3c769f92a1...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_TabGroups.tsv	aaffd5f79baf...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_TabIndividuals.tsv	e3daf0730eae...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Frank2009/.Frank2009_TabInterventions.tsv	4286526be1d0...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Gaedigk2013/.Gaedigk2013_Fig1.tsv	81d26e93ffbd...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Gaedigk2013/.Gaedigk2013_TabGroups.tsv	f83738fa5a2b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Gaedigk2013/.Gaedigk2013_TabInterventions.tsv	de7b42be55d3...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Gaedigk2018/.Gaedigk2018_Fig1.tsv	d2b56af4da16...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Gaedigk2018/.Gaedigk2018_TabIndividuals.tsv	4d2dddc23173...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Gaedigk2018/.Gaedigk2018_TabInterventions.tsv	143c653b5d69...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Gazzaz2018/.Gazzaz2018_Fig3.tsv	b80b3ae79948...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Gazzaz2018/.Gazzaz2018_Fig3_median.tsv	f5e21455d2ac...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Gazzaz2018/.Gazzaz2018_Tab4.tsv	d887f32b0ecb...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_Fig1.tsv	ef112f491f1e...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_Fig1A.tsv	3cf8f2e5a82d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_Fig1B.tsv	de290d46002d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_Fig2.tsv	5b20cff38be4...

File	SHA-256
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_Fig3.tsv	248485683918...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_Fig5.tsv	f35b5d9b9eb1...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_Fig7.tsv	0b91962977a1...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_Fig8.tsv	236262652de2...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_TabA.tsv	8d9bd3d54340...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_TabData.tsv	9b5320a9466f...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_TabGroups.tsv	38d1411b4735...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_TabIndividuals.tsv	4b651c53df06...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_TabInterventions.tsv	09d7a1414a2b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1991/.Hou1991_TabText.tsv	c8b6a40c51e3...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1996/.Hou1996_Fig1.tsv	3096cd8476e6...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1996/.Hou1996_Fig2.tsv	bc5a1624ef58...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1996/.Hou1996_Fig3.tsv	3096cd8476e6...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1996/.Hou1996_Tab1.tsv	ba7512cc1e24...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1996/.Hou1996_TabGroups.tsv	1efd29832580...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1996/.Hou1996_TabIndividuals.tsv	27fc8ed73b9b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1996/.Hou1996_TabInterventions.tsv	d15ab12d4424...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hou1996/.Hou1996_TabText.tsv	113c8fc28ed1...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hu2011/.Hu2011_Fig6.tsv	69317f0ff8c3...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Hu2011/.Hu2011_Tab4.tsv	2e5ac5d25bb0...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Jones1996/.Jones1996_Fig2.tsv	a93bc0765d09...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Jones1996/.Jones1996_Fig2_group.tsv	1fb7ccf2c634...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Jones1996/.Jones1996_Fig3.tsv	c70312b39cae...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Jones1996/.Jones1996_Fig3_group.tsv	fe4dda31120d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Jones1996/.Jones1996_Fig4.tsv	f66ea9d55ddf...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Jones1996/.Jones1996_Fig5.tsv	b1a3080340dd...

File	SHA-256
pkdb_models/models/dextromethorphan/data/dextromethorphan/Jones1996/.Jones1996_Tab1.tsv	9924e4507a56...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Jones1996/.Jones1996_Tab1_group.tsv	cd47f288998a...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Jones1996/.Jones1996_Tab2.tsv	453d30942ea4...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Jones1996/.Jones1996_Tab2_group.tsv	2e7b315bc1fc...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Jones1996/.Jones1996_TabA.tsv	70bbe290fe14...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Jones1996/.Jones1996_TabIndividuals.tsv	70bbe290fe14...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Kazis2009/.Kazis2009_Fig1.tsv	fca7ad57ecc5...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Kazis2009/.Kazis2009_Fig2.tsv	54ab11d5e461...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Kazis2009/.Kazis2009_Tab1.tsv	7f746890aea2...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Kazis2009/.Kazis2009_Tab2.tsv	1b9791fd1c40...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Kazis2009/.Kazis2009_Tab3.tsv	6b5c1d78b654...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Kazis2009/.Kazis2009_Tab4.tsv	a9813b316d49...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Kazis2009/.Kazis2009_TabIndividuals.tsv	ce5b235e5d60...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Kazis2009/.Kazis2009_TabInterventions.tsv	15dbcc510eb1...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Kim2016/.Kim2016_Fig1.tsv	c44528ec99d6...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Kim2016/.Kim2016_Tab2.tsv	ebf3972b7a8d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Koehler1997/.Koehler1997_Fig2.tsv	672e0c08d855...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Koehler1997/.Koehler1997_Fig3.tsv	7a75656f45ee...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Koehler1997/.Koehler1997_Tab1.tsv	f5d9a5d80fbd...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Koehler1997/.Koehler1997_Tab1_metabolic_ratio.tsv	0a6865d16f4a...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Koehler1997/.Koehler1997_Tab2.tsv	c2bef2dd7523...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Koehler1997/.Koehler1997_Tab2_metabolic_ratio.tsv	0fbceacb7128...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Koehler1997/.Koehler1997_TabData.tsv	88891f5719fd...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Koehler1997/.Koehler1997_TabGroup.tsv	b24e79972173...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Koehler1997/.Koehler1997_TabIndividuals.tsv	3ec9a18b5c27...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Lenuzza2016/.Lenuzza2016_Fig1.tsv	4c28ca73ca51...

File	SHA-256
pkdb_models/models/dextromethorphan/data/dextromethorphan/Lenuzza2016/.Lenuzza2016_Fig2.tsv	9d6e6f0d9fb0...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Lenuzza2016/.Lenuzza2016_TabIndividuals.tsv	6b984558f57d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Loetsch2009/.Loetsch2009_Fig2.tsv	dc63b566f7db...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Loetsch2009/.Loetsch2009_Tab2.tsv	412be5eaaf4e...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Lopez2005/.Lopez2005_Fig1.tsv	f38cca0a92cd...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Lopez2005/.Lopez2005_Fig2.tsv	ecd5ed849fe0...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Lopez2005/.Lopez2005_TabGroups.tsv	036ce9d42119...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Lopez2005/.Lopez2005_TabInterventions.tsv	11ecee23a773...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Montane2013/.Montane2013_Fig1.tsv	f38cca0a92cd...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Montane2013/.Montane2013_Fig2.tsv	a4d0a24930e0...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Montane2013/.Montane2013_Fig3.tsv	ae6913946ad2...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Montane2013/.Montane2013_TabData.tsv	ee83dced4d50...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Montane2013/.Montane2013_TabGroups.tsv	1944915ab7fe...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Montane2013/.Montane2013_TabInterventions.tsv	d47203d3a41c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Myrand2008/.Myrand2008_Fig1.tsv	f38cca0a92cd...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Myrand2008/.Myrand2008_Fig2.tsv	ee2a5b7a6f83...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Myrand2008/.Myrand2008_Fig5.tsv	10b15de3ec2a...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Myrand2008/.Myrand2008_Fig6.tsv	17ffbf5b977d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Myrand2008/.Myrand2008_TabGroups.tsv	825ff7b5c516...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Myrand2008/.Myrand2008_TabInterventions.tsv	d430963f971e...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nagai1996/.Nagai1996_Fig1.tsv	3967b5de0429...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nagai1996/.Nagai1996_Fig2.tsv	75b70094fab9...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nagai1996/.Nagai1996_Fig3.tsv	a5ec85f6264b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nagai1996/.Nagai1996_Fig4.tsv	92d434bb7965...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nagai1996/.Nagai1996_Tab1.tsv	176ef8db6e42...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nagai1996/.Nagai1996_Tab1_group.tsv	0b7cb56f5f48...

File	SHA-256
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nagai1996/.Nagai1996_Tab2.tsv	f1ef573b8aee...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nagai1996/.Nagai1996_Tab2_group.tsv	9836bff8eac6...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nagai1996/.Nagai1996_Tab3.tsv	b115a0fcaf7e...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nagai1996/.Nagai1996_TabIndividuals.tsv	656545197a69...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nakashima2007/.Nakashima2007_Fig2.tsv	4a5ffbf82af8...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nakashima2007/.Nakashima2007_Tab1.tsv	8d8b38984a6e...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nakashima2007/.Nakashima2007_Tab2.tsv	5c8207edd58e...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nyunt2008/.Nyunt2008_Fig1.tsv	d6666c142221...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Nyunt2008/.Nyunt2008_Tab1.tsv	8344d5042e5b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/ONeil2000/.ONeil2000_Fig2.tsv	db5ceeee580a...
pkdb_models/models/dextromethorphan/data/dextromethorphan/ONeil2000/.ONeil2000_Tab1.tsv	f3417a0dcb4d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Pope2004/.Pope2004_Fig1.tsv	8162c25d1e26...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Pope2004/.Pope2004_Fig2.tsv	3492663f79f9...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Pope2004/.Pope2004_Fig3.tsv	42a5e30fb06e...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Pope2004/.Pope2004_Tab2.tsv	926d757f1c0c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Pope2004/.Pope2004_Tab3.tsv	a09d87c5e4f5...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Pope2004/.Pope2004_Tab4.tsv	b5815f7546c4...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Pope2004/.Pope2004_Tab5.tsv	f97e145161be...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Pope2004/.Pope2004_TabA.tsv	510727d67842...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Qiu2016/.Qiu2016_Fig1.tsv	e94d0d19414a...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Qiu2016/.Qiu2016_Fig2.tsv	fae9efca0f0d...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Qiu2016/.Qiu2016_Fig3.tsv	d9c7994ec5ac...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Qiu2016/.Qiu2016_Tab1.tsv	5fa9448f57ff...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Qiu2016/.Qiu2016_TabIndividuals.tsv	280351775a4a...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schadel1995/.Schadel1995_Fig2.tsv	643b0c7461d4...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schadel1995/.Schadel1995_Fig3.tsv	7fe119e0cdd2...

File	SHA-256
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schadel1995/.Schadel1995_Fig4.tsv	2db678fc717f...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schadel1995/.Schadel1995_Tab2.tsv	a827c400be43...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schadel1995/.Schadel1995_Tab2_group.tsv	25f68bb81f8f...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schadel1995/.Schadel1995_Tab3.tsv	b43180fa66f3...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schadel1995/.Schadel1995_TabIndividuals.tsv	395a2e0c6c99...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schadel1995/.Schadel1995_Tab_individuals.tsv	395a2e0c6c99...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schadel1995/study.json	11fe052943e1...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schoedel2012/.Schoedel2012_Fig3.tsv	e3770a8d7abf...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schoedel2012/.Schoedel2012_Fig4.tsv	64edcc84a606...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schoedel2012/.Schoedel2012_Tab1.tsv	c8d640e49978...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schoedel2012/.Schoedel2012_Tab2.tsv	a16c5d1a882a...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schoedel2012/.Schoedel2012_Tab3.tsv	8a1125d68ec2...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Schoedel2012/.Schoedel2012_TabGroups.tsv	658b0d9532cc...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Streetman1999/.Streetman1999_Fig1.tsv	d7fd12716c4f...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Tamminga2001/.Tamminga2001_Fig1.tsv	6fc76165fceb...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Tamminga2001/.Tamminga2001_TabInterventions.tsv	a78d51311b49...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Tamminga2001b/.Tamminga2001b_Fig1.tsv	e15f952f8792...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Tamminga2001b/.Tamminga2001b_TabGroups.tsv	7275e5f18c54...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Tamminga2001b/.Tamminga2001b_TabIndividuals.tsv	3ba6f866bf08...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Tamminga2001b/.Tamminga2001b_TabInterventions.tsv	70c54adfcf13...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Tenneze1999/.Tenneze1999_Fig1.tsv	4b14e7a932a8...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Tenneze1999/.Tenneze1999_Fig2.tsv	45803891ec0b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Tenneze1999/.Tenneze1999_Tab2.tsv	07e41cbfa293...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Tenneze1999/.Tenneze1999_TabGroups.tsv	5b1c0ddf346e...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Tenneze1999/.Tenneze1999_TabIndividuals.tsv	0ee76e6b05b1...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Tenneze1999/.Tenneze1999_TabInterventions.tsv	3dd97bafe358...

File	SHA-256
pkdb_models/models/dextromethorphan/data/dextromethorphan/Wyen2008/.Wyen2008_Fig1.tsv	2a51bf9cceea...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Wyen2008/.Wyen2008_Fig2.tsv	7cc2ece62e66...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Wyen2008/.Wyen2008_Fig3.tsv	a2c6f39f3fb2...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Wyen2008/.Wyen2008_Fig4.tsv	2f7146f74f46...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Wyen2008/.Wyen2008_Fig5.tsv	cf7337424aee...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Wyen2008/.Wyen2008_Fig6.tsv	936f85d045fa...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Wyen2008/.Wyen2008_Tab1.tsv	eb960e8f359c...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Yeh2003/.Yeh2003_Fig1.tsv	52577df39966...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Yeh2003/.Yeh2003_Tab1.tsv	02717da8dbc7...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Yeh2003/.Yeh2003_Tab1_Group.tsv	47be7ab91566...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Yeh2003/.Yeh2003_Tab2.tsv	7737c62fdb94...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Yeh2003/.Yeh2003_Tab2_Group.tsv	cf9685b6f9b8...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Yeh2003/.Yeh2003_TabIndividuals.tsv	54d1cb977e0b...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Yeh2003/.Yeh2003_TabInterventions.tsv	d6fa3153757a...
pkdb_models/models/dextromethorphan/data/dextromethorphan/Zawertailo2010/.Zawertailo2010_Fig1.tsv	95cd56fcc94c...
pkdb_models/models/dextromethorphan/models/output/models/dextromethorphan_body_flat.xml	84e5b0f6b7c5...

2 Results: Dextromethorphan PBPK model: fit PK

2.1 Parameters

Table 2: Parameters of the fit PK_20261020_094137__4239e, the parameter set PK_20261020_094137__4239e_0, cost 309, with their bounds in the optimization. The 95% confidence intervals are those of the profile likelihood, LB and UB an interval which is open towards the lower or the upper bound, and the Wald intervals of the Fisher information for a parameter without a profile. Identifiable is the classification of the profile: yes, no (LB) or no (UB) if the profile does not reach the threshold towards the lower or the upper bound, no if it reaches it on neither side, structural if it is flat.

Parameter	Description	Value [95% CI]	Unit	Bounds	Identifiable
Ka_dis_dxm	Ka_dis [1/hr] dissolution dextromethorphan	0.402 [0.339, 0.476]	1/hr	[0.05, 10]	yes
GU__F_dxm_abs	fraction absorbed dextromethorphan	0.524 [0.469, 0.581]	dimensionless	[0.3, 1]	yes
LI__DXMCYP2D6_Vmax	Vmax DXMCYP2D6	0.00591 [0.00291, 0.00956]	mmol/min/l	[0.0001, 1]	yes

Parameter	Description	Value [95% CI]	Unit	Bounds	Identifiable
LI__DXMCYPX_Vmax	Vmax DXMCYPX	0.000455 [0.000285, 0.000669]	mmol/min/l	[1e-06, 0.01]	yes
LI__DXMNDM_Vmax	Vmax DXMNDM	0.00117 [0.000421, 0.00218]	mmol/min/l	[1e-05, 1]	yes
LI__DXONDM_Vmax	Vmax DXONDM	0.322 [0.264, 0.393]	mmol/min/l	[0.001, 50]	yes
LI__DXOUGT_Vmax	Vmax DXOUGT	0.637 [0.529, 0.768]	mmol/min/l	[0.01, 100]	yes
Kp_dxm	tissue/plasma partition coefficient dxm	9.49 [6.66, 16.5]	dimensionless	[1, 20]	yes
Kp_mm	tissue/plasma partition coefficient mm	0.0933 [0.0225, 0.216]	dimensionless	[0.02, 3]	yes
KI__DXMEX_f	renal clearance of unbound dextromethorphan relative to the GFR	0.656 [0.482, 0.856]	dimensionless	[0.05, 10]	yes
KI__DXOEX_f	renal clearance of unbound dextrorphan relative to the GFR	1.87 [1.44, 2.39]	dimensionless	[0.05, 50]	yes
KI__DXOGLUEX_f	renal clearance of unbound dextrorphan glucuronide relative to the GFR	2.27 [1.87, 2.74]	dimensionless	[0.05, 20]	yes
KI__MMEX_f	renal clearance of unbound N-demethylated morphinans relative to the GFR	1.66 [1.13, 2.47]	dimensionless	[0.05, 20]	yes

2.2 Data

Table 3: Data of the fit PK_20261020_094137__4239e: the number of fit mappings and, in parentheses, of data points per study and kind. The training data enters the cost, the validation data and the outliers are evaluated only.

Study	Training	Validation	Outlier	Total
Abdelrahman1999	7 (9)	7 (9)	-	14 (18)
Armani2017	-	2 (26)	-	2 (26)
Barnhart1980	5 (74)	3 (30)	-	8 (104)
Capon1996	2 (23)	1 (11)	2 (18)	5 (52)
Chen2017	8 (442)	-	-	8 (442)
Chladek2000	5 (243)	1 (6)	-	6 (249)
Demirbas1998	-	4 (60)	-	4 (60)
Dorado2017	-	30 (161)	-	30 (161)
Doroshyenko2013	1 (11)	1 (12)	-	2 (23)
Duedahl2005	-	1 (19)	-	1 (19)
Dumond2010	-	3 (30)	-	3 (30)
Edwards2017	-	4 (56)	-	4 (56)
Eichhold1997	2 (18)	-	-	2 (18)
Eichhold2007	4 (58)	4 (57)	-	8 (115)

Study	Training	Validation	Outlier	Total
Frank2009	116 (555)	122 (581)	74 (369)	312 (1505)
Gaedigk2013	7 (110)	-	-	7 (110)
Hou1991	-	6 (80)	-	6 (80)
Hu2011	-	2 (22)	-	2 (22)
Jones1996	18 (138)	7 (49)	3 (21)	28 (208)
Koehler1997	6 (28)	2 (10)	-	8 (38)
Lenuzza2016	2 (27)	18 (204)	-	20 (231)
Lopez2005	-	16 (88)	-	16 (88)
Montane2013	-	48 (222)	-	48 (222)
Myrand2008	-	34 (413)	-	34 (413)
Nagai1996	6 (64)	7 (42)	-	13 (106)
Nakashima2007	2 (16)	2 (19)	-	4 (35)
Nyunt2008	1 (10)	-	-	1 (10)
Oh2012	-	2 (28)	-	2 (28)
Pope2004	-	6 (50)	-	6 (50)
Qiu2016	3 (42)	3 (42)	-	6 (84)
Schadel1995	14 (57)	9 (45)	-	23 (102)
Schoedel2012	-	2 (18)	-	2 (18)
Tamminga2001b	9 (191)	-	-	9 (191)
Tenneze1999	5 (21)	5 (20)	-	10 (41)
Yamazaki2016	-	6 (88)	-	6 (88)
Zawertailo2010	4 (28)	4 (28)	-	8 (56)
Total	227 (2165)	362 (2526)	79 (408)	668 (5099)

2.3 Goodness of fit

Table 4: Goodness of fit of the fit PK_20261020_094137__4239e per kind of data, the parameter set PK_20261020_094137__4239e_0. The cost is defined on the training data only, 2-fold is the percentage of the data points with a positive value and prediction whose prediction is within twofold of the data.

Kind	n	cost	MSE	RMSE	NRMSE	RMSE_w	R^2	2-fold [%]	AIC	BIC
training	2165	309	7.05	2.66	1.62	0.534	0.079	46.9	4255.1	4329.0
validation	2526	-	98	9.9	8.99	7.18	−0.149	42.0	11607.6	11683.5
outlier	408	-	0.0444	0.211	2.11	1.67	0.635	37.5	−1244.7	−1192.5

2.4 Figures

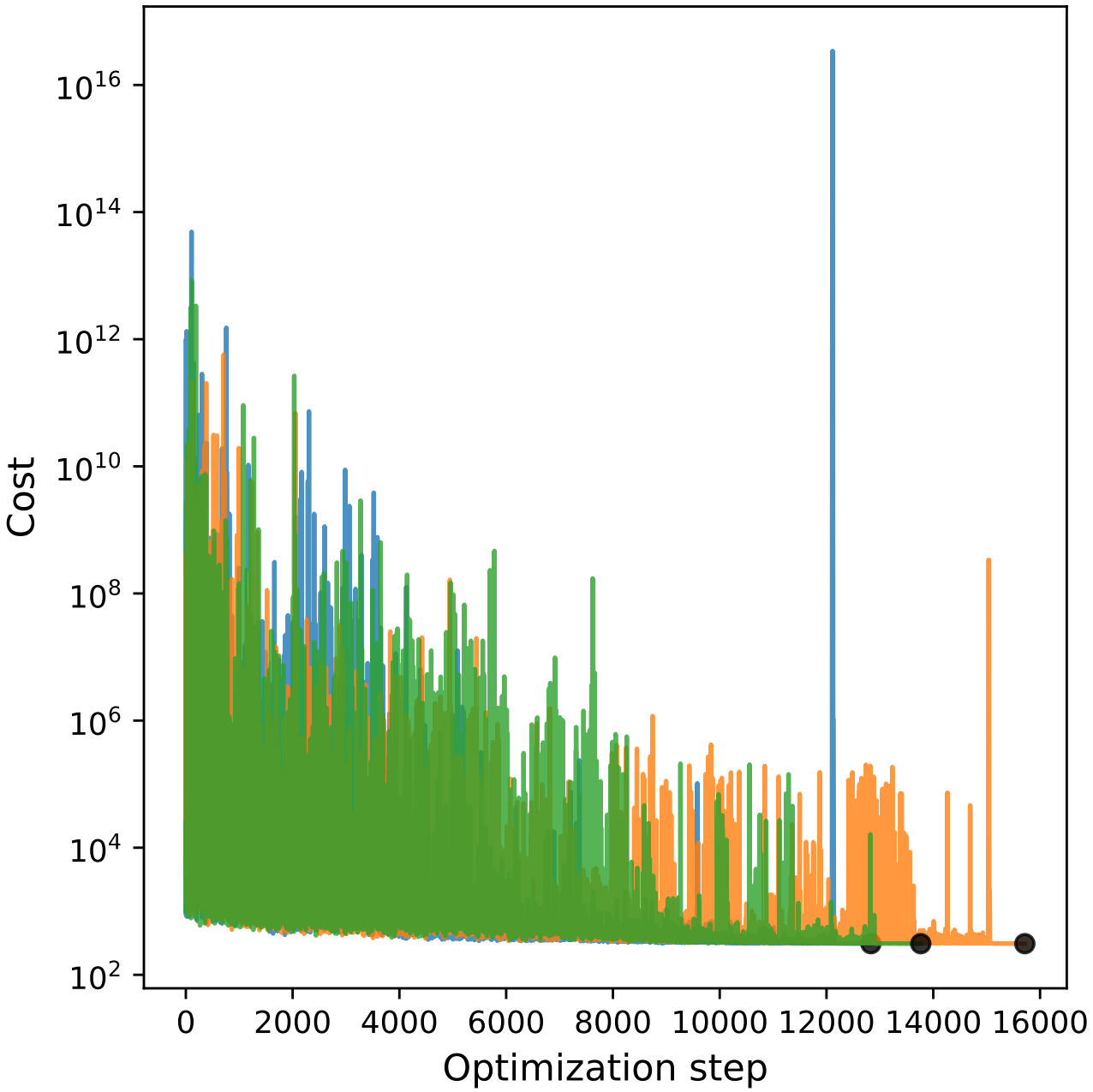


Figure 1: Cost of the optimizers over their steps, fit PK_20261020_094137__4239e. The cost of every optimization run over its steps. Runs which end at the same cost found the same optimum; a run which stops much higher is stuck in a local minimum.

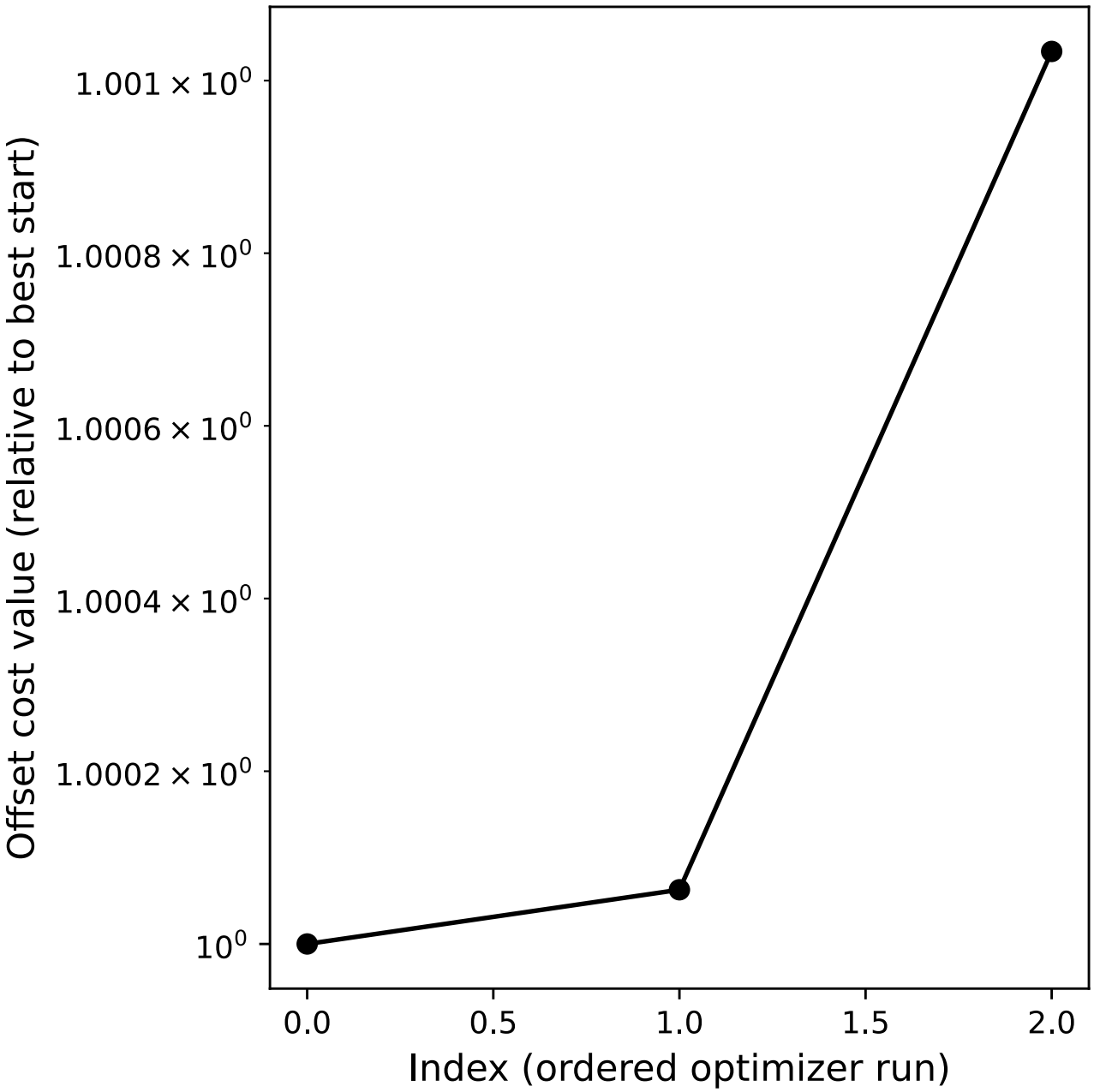


Figure 2: Cost of the optimization runs, ordered, fit PK_20261020_094137__4239e. The final cost of the runs, ordered. A flat plateau at the left is the global optimum found repeatedly, which is the evidence that the multistart converged; steps to the right are local minima.

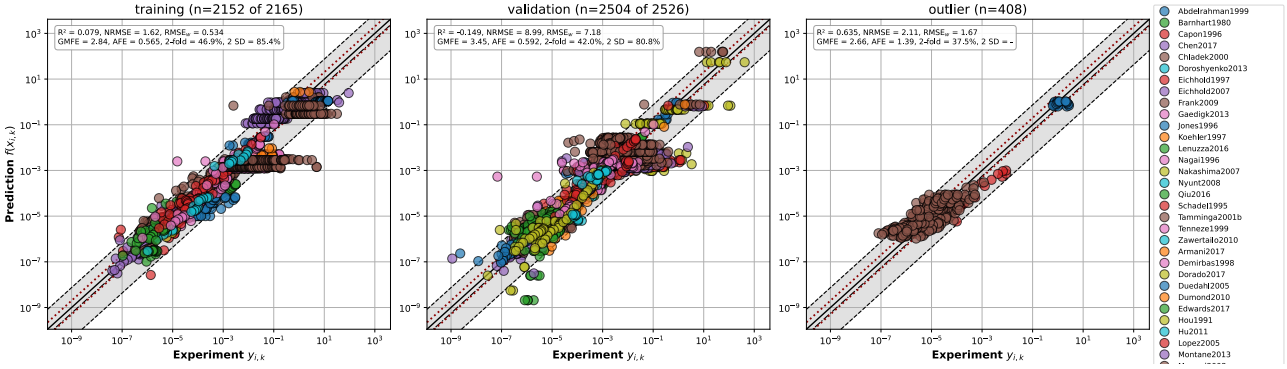


Figure 3: Prediction against the measured data points, per kind of fit mapping, with the agreement of the training data, the lines of twofold and the metrics of the panel, fit PK_20261020_094137__4239e. Within twofold of the data are 1010 of the 2152 training data points (46.9%), 1051 of the 2504 validation data points (42.0%) and 153 of the 408 outlier data points (37.5%); only the data points with a positive value and prediction are counted. Prediction against measurement, one point per data point, with a panel per kind of fit mapping. The points scatter around the diagonal when the model describes the data; a systematic deviation from it is a systematic error of the model. The band is the agreement of the training data, i.e. the same band the Bland-Altman plot draws, here as lines parallel to the diagonal. The dark red dotted lines are twofold, prediction = 2 x measurement and prediction = measurement / 2; the caption states the points between them of every kind. A point which is zero or negative is not drawn on the logarithmic axes and not counted, the title of a panel then states the points drawn of all. The validation panel shows how the fit describes data it was not fitted on, the outlier panel the data the fit dropped. The box of a panel carries its R^2, its normalized RMSE, the RMSE of the residuals of the cost, its fold errors GMFE and AFE and the percentages of its points within twofold and within two standard deviations of the data.

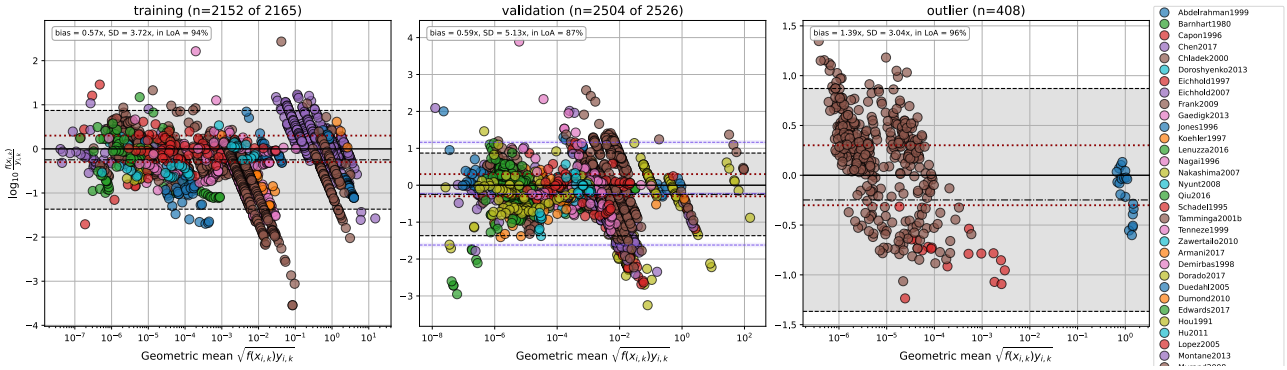


Figure 4: Agreement of prediction and measurement as a ratio, per kind of fit mapping, with the agreement of the training data and the agreement of the panel, fit PK_20261020_094137__4239e. The ratio of prediction and measurement over the geometric mean of the two, i.e. the goodness of fit with the diagonal turned into the horizontal. The band is the same, the bias and the limits of agreement `bias ± 1.96 SD` of the training data as fold factors. A bias away from 1 is a systematic over- or underprediction, wide limits are a large scatter, and a trend over the mean is a model which describes the large or the small values better. The box of a panel carries the bias and the SD of its own points as fold factors and the share of them inside the limits of agreement of the training data.

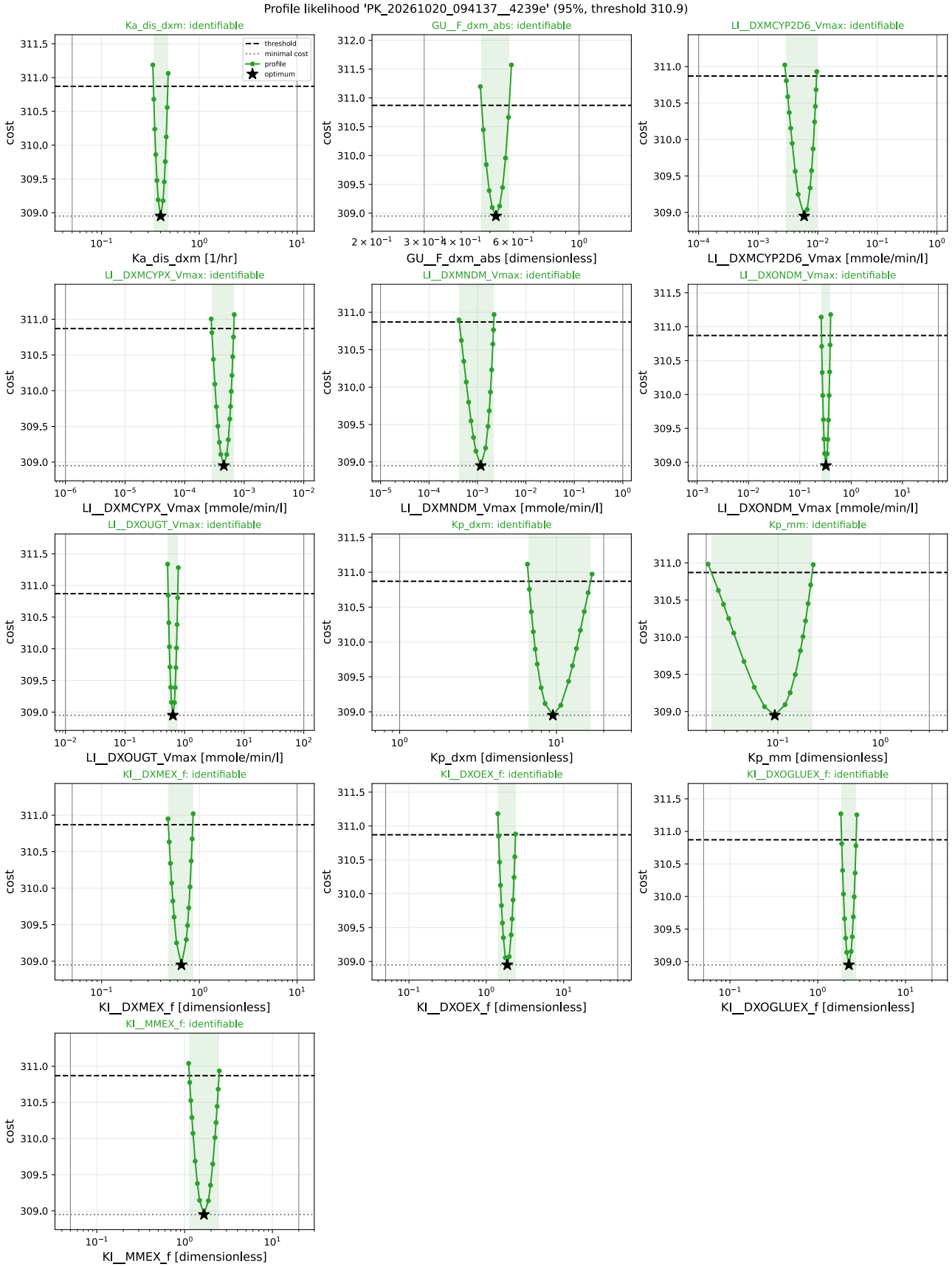


Figure 5: Profile likelihood of every parameter: the cost with the parameter fixed and the other parameters optimized, the threshold of the confidence level and the confidence interval, fit PK_20261020_094137__4239e. One profile per parameter. A profile which crosses the dashed threshold on both sides gives a finite confidence interval; one which stays below it up to a bound is practically non-identifiable; a flat profile is structurally non-identifiable, the parameter is compensated by the others.

2.5 Provenance

Table 5: Provenance of the fit PK_20261020_094137__4239e: the versions it ran with and the revision of its models and data; the SHA-256 checksums of the models and data are listed in Table 1.

Item	Value
Fit	`PK_20261020_094137__4239e`
Created	2026-10-20 07:41:49 UTC
Seed	1234
Python	3.14.4
sbmlsim	0.10.0
libRoadRunner	2.10.0
SciPy	1.18.1
NumPy	2.5.3
Repository	`/home/mkoenig/git/pkdb_models-dextromethorphan`
Commit	`aa8deb245665a611f7c5be43a35ced599c893cf3`
Uncommitted changes	no

Bibliography

[1] M. König, "sbmlsim: SBML simulation made easy." Zenodo, 2026. doi: 10.5281/zenodo.3597770. Version 0.10.0.

[2] R. Storn and K. Price, "Differential evolution - a simple and efficient heuristic for global optimization over continuous spaces," *Journal of Global Optimization*, vol. 11, pp. 341–359, 1997, doi: 10.1023/A:1008202821328.

[3] P. Virtanen *et al.*, "SciPy 1.0: fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020, doi: 10.1038/s41592-019-0686-2. Version 1.18.1.

[4] M. D. McKay, R. J. Beckman, and W. J. Conover, "A comparison of three methods for selecting values of input variables in the analysis of output from a computer code," *Technometrics*, vol. 21, no. 2, pp. 239–245, 1979, doi: 10.2307/1268522.

[5] C. Welsh, J. Xu, L. Smith, M. König, K. Choi, and H. M. Sauro, "libRoadRunner 2.0: a high performance SBML simulation and analysis library," *Bioinformatics*, vol. 39, no. 1, p. btac770, 2023, doi: 10.1093/bioinformatics/btac770. Version 2.10.0.

[6] A. C. Hindmarsh *et al.*, "SUNDIALS: suite of nonlinear and differential/algebraic equation solvers," *ACM Transactions on Mathematical Software*, vol. 31, no. 3, pp. 363–396, 2005, doi: 10.1145/1089014.1089020.

[7] H. Akaike, "A new look at the statistical model identification," *IEEE Transactions on Automatic Control*, vol. 19, no. 6, pp. 716–723, 1974, doi: 10.1109/TAC.1974.1100705.

[8] G. Schwarz, "Estimating the dimension of a model," *The Annals of Statistics*, vol. 6, no. 2, pp. 461–464, 1978, doi: 10.1214/aos/1176344136.

[9] J. M. Bland and D. G. Altman, "Statistical methods for assessing agreement between two methods of clinical measurement," *The Lancet*, vol. 327, no. 8476, pp. 307–310, 1986, doi: 10.1016/S0140-6736(86)90837-8.

[10] A. Raue *et al.*, "Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood," *Bioinformatics*, vol. 25, no. 15, pp. 1923–1929, 2009, doi: 10.1093/bioinformatics/btp358.

[11] C. Kreutz, A. Raue, D. Kaschek, and J. Timmer, "Profile likelihood in systems biology," *The FEBS Journal*, vol. 280, no. 11, pp. 2564–2571, 2013, doi: 10.1111/febs.12276.

[12] F.-G. Wieland, A. L. Hauber, M. Rosenblatt, C. Tönsing, and J. Timmer, "On structural and practical identifiability," *Current Opinion in Systems Biology*, vol. 25, pp. 60–69, 2021, doi: 10.1016/j.coisb.2021.03.005.

[13] C. R. Harris *et al.*, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020, doi: 10.1038/s41586-020-2649-2. Version 2.5.3.